Personalized Travel Itinerary Planner: AI-Powered Travel Experience Engine

Project Description:

The Personalized Travel Itinerary Planner harnesses Generative AI and geospatial optimization to provide intelligent, hyper-personalized travel planning, transforming abstract travel desires ("chill sunsets", "high-adrenaline adventure", "Michelin street foodie", "deep cultural immersion") into day-by-day, geographically optimized, and budget-conscious travel itineraries.
The platform includes:

- Vibe & Preference Engine: Evaluates traveler pace, budget caps, companion style, and niche aesthetic vibes with multi-dimensional vector compatibility scoring.
- AI Itinerary Generator: Powered by Google Gemini 2.5 Flash, synthesizing structured morning-to-night schedules with duration counters, estimated activity costs, and local insider tips.
- Route & Time Optimizer: Employs the Haversine formula and a 2-opt TSP (Traveling Salesperson Problem) heuristic algorithm to eliminate zigzagging and minimize daily transit time.
- Dynamic Vibe Sliders: Allows on-the-fly itinerary recalibration (+40% Adventure, Sunset & Chill Mode, Budget-friendly swaps).
- Multi-Format Export Subsystem: 1-click sync to Apple/Google/Outlook Calendars (.ics), luxury printable PDF dossier, and raw JSON backup.

Built with a high-performance FastAPI (Python 3.11) backend, a luxury glassmorphism React 18 + Vite frontend, and persistent Supabase PostgreSQL cloud storage, the platform is publicly deployed on Vercel (Frontend) and Render (Backend).

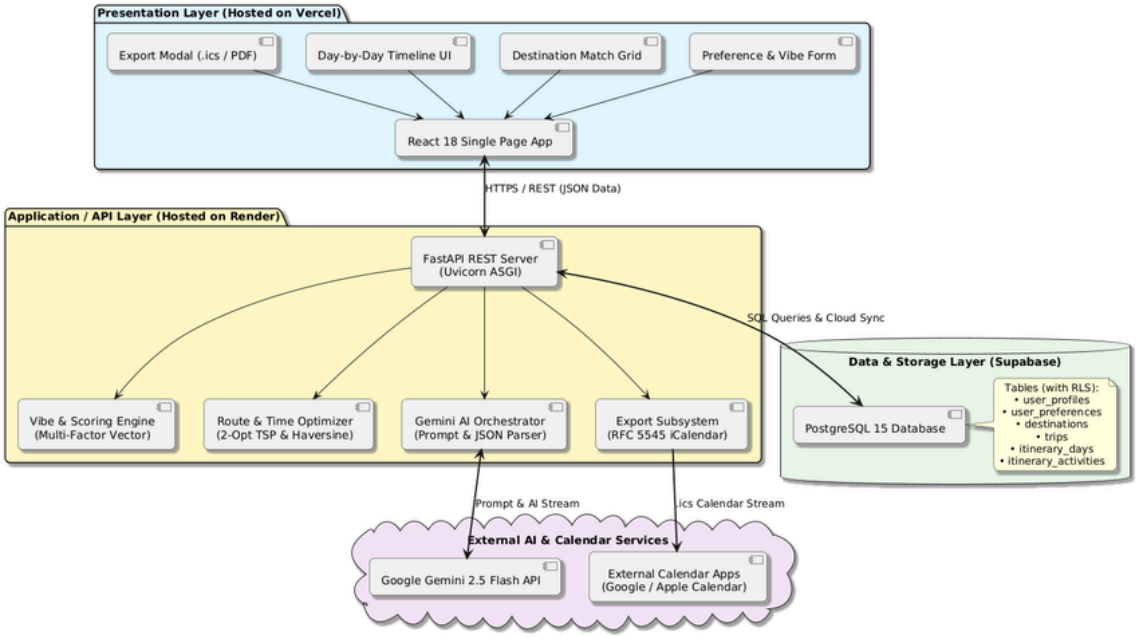
Scenarios

Scenario 1: The Busy Professional Planning a Solo Chill Getaway
A user has 3 days in Kyoto and selects a Relaxed pace with a Sunset & Chill vibe and moderate budget. The AI generates a serene morning-to-evening schedule featuring quiet temple gardens, canal-side matcha cafes, and sunset viewpoint spots. The Route Optimizer groups nearby Higashiyama spots together, saving 2+ hours of transit. The traveler syncs the schedule directly into their Google Calendar via .ics.

Scenario 2: High-Adrenaline Adventure Backpackers on a Strict Budget
A group of college friends planning a 4-day trip to Queenstown, New Zealand selects Fast-Paced, Adventure, and a strict budget cap of $80/day. The AI schedules bungee jumping, mountain biking trails, and scenic hikes, calculating activity entrance fees and local transport costs. When the group wants more downtime on Day 3, they slide the dynamic vibe slider to +30% Chill, instantly swapping a strenuous climb for a lakeside lounge.

Scenario 3: Culture & Michelin Street Food Enthusiast
A solo foodie visiting Rome selects Moderate pace, Cultural Immersion, and Foodie Vibe. The recommendation engine highlights Rome with a 96% compatibility match. The generated itinerary sequences morning historic monument visits with authentic local trastevere food tastings, ensuring opening hours align and transit backtracking is eliminated. The user exports a printable luxury PDF dossier for offline reference.

Technical Architecture:



Architecture Description:
The system utilizes a decoupled, modern three-tier cloud architecture:

1. Frontend Layer (Vercel): Single Page Application built with React 18, Vite, and bespoke Vanilla CSS implementing a responsive dark glassmorphism design system.
2. API & Logic Layer (Render): FastAPI asynchronous server orchestrating Gemini prompt engineering, mathematical route optimization, validation via Pydantic schemas, and CORS security.
3. Database Layer (Supabase): PostgreSQL 15 relational storage enforcing Row-Level Security (RLS) policies across user profiles, destination catalogs, and itinerary timelines.

Database Entity-Relationship (ER) Overview:

- user_profiles (id, email, full_name, created_at)
- user_preferences (id, user_id, primary_vibe, secondary_vibes, travel_style, pace, daily_budget_usd)
- destinations (id, name, country, vibes, avg_daily_cost, latitude, longitude, rating, highlights)
- trips (id, user_id, title, destination_name, start_date, end_date, total_days, total_budget, vibe_style)
- itinerary_days (id, trip_id, day_number, date, theme_title, estimated_daily_cost)
- itinerary_activities (id, day_id, trip_id, time_slot, start_time, title, location_name, latitude, longitude, estimated_cost, duration_minutes, order_index)

Pre-requisites:

- FastAPI Framework Knowledge: FastAPI Documentation
- Google Gemini API Familiarity: Google AI Studio Documentation
- React & Vite Skills: React Docs & Vite Docs
- Supabase & PostgreSQL Proficiency: Supabase Documentation
- Python 3.10+ & Node.js 18+ Proficiency
- Git & GitHub Version Control
- Cloud Deployment: Render Docs & Vercel Docs

Project Workflow:

Milestone 1: Model Selection and Architecture

- Activity 1.1: Configure Google Gemini API Key: Generate Gemini API key from Google AI Studio and configure .env securely.
- Activity 1.2: Model Evaluation: Select Gemini 2.5 Flash for its low latency (< 2.5s), high reasoning capability, and deterministic JSON generation.
- Activity 1.3: Define System Architecture: Draft component diagrams, ER models, and RESTful route schemas.
- Activity 1.4: Set Up Development Environment: Initialize virtual environment, install FastAPI dependencies, configure React Vite frontend.

Milestone 2: Core Functionalities Development

- Activity 2.1: Vibe & Recommendation Engine: Vector scoring calculating destination compatibility percentage.
- Activity 2.2: Daily Itinerary Builder: Multi-slot daily schedule planner (Morning, Afternoon, Evening, Night).
- Activity 2.3: 2-Opt TSP Route Optimizer: Python implementation using Haversine formula to eliminate transit backtracking.

Milestone 3: Backend API Development (main.py & Services)

- Activity 3.1: REST Endpoints: Build /api/recommendations, /api/itineraries/generate, /api/itineraries/optimize, and /api/export/{trip_id}.
- Activity 3.2: Robust JSON Extraction: Implement regex and code-fence sanitizers for flawless LLM response parsing.

Milestone 4: Frontend Development

- Activity 4.1: Glassmorphism UI: Design responsive, dark-mode luxury theme using modern CSS variables and typography.
- Activity 4.2: Dynamic Components: Build PreferenceForm.jsx, RecommendationGrid.jsx, ItineraryTimeline.jsx, dynamic vibe sliders, and export modals.

Milestone 5: Deployment

- Activity 5.1: Backend on Render: Deploy Python FastAPI web service with environment variables and Uvicorn runtime.
- Activity 5.2: Frontend on Vercel: Connect GitHub repository to Vercel for zero-config global Edge CDN hosting.
- Activity 5.3: Verification: Validate live cross-origin requests, database persistence, and calendar export downloads.

Milestone 6: Conclusion

Summarize project outcomes, user experience improvements, system scalability, and future enhancements.

Exploring the Website's Web Pages:

Conclusion:

The Personalized Travel Itinerary Planner successfully demonstrates how Generative AI combined with algorithmic geospatial optimization can solve real-world travel planning friction. By replacing hours of manual research with tailored, geographically logical itineraries in seconds, the platform empowers travelers with confident, flexible, and stress-free vacation planning. Future enhancements include live flight/hotel API integrations and collaborative group trip editing.